→ Naive Bayes Algorithm:

Naive Bayes classifier:

# Spam email classifier

Dataset

| $x_1$ | $x_2$ | $x_3$ | Label |
|---|---|---|---|
| 1 | 0 | 1 | spam |
| 1 | 1 | 0 | spam |
| 1 | 1 | 0 | spam |
| 0 | 0 | 1 | No spam |
| 0 | 0 | 0 | " |
| 1 | 1 | 1 | " |
| 0 | 1 | 0 | spam |
| 1 | 0 | 0 | " |
| 0 | 1 | 0 | " |
| 1 | 0 | 0 | " |

Test data

1 1 0 - spam or No spam

→ 2/7 → spam.

Vector for all word in english and is encoded.

→ "Embedding vector"

Dictionary.

$$\begin{array}{c} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x^d \end{array} \begin{bmatrix} a & 1 \\ & 0 \\ & 0 \\ & 1 \\ & \vdots \end{bmatrix} \begin{array}{l} a \\ \text{back} \\ \text{you} \\ \\ \\ \end{array}$$

Big Master Joint distribution Table $y$ = spam = 1

$y$ = No spam = 0.

| $x_1$ | $x_2$ | $x_3$ | $P(y=1)$ | $P(y=0)$ | |
|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1/3 | 1 out of 3 No spams |
| 0 | 0 | 1 | 0 | 1/3 | |
| 0 | 1 | 0 | 2/7 | 0 | as 2 out of 7. |
| 0 | 1 | 1 | 0 | 0 | → not available. |
| 1 | 0 | 0 | 2/7 | 0 | |
| 1 | 0 | 1 | 1/7 | 0 | |
| 1 | 1 | 0 | 2/7 | 0 | |
| 1 | 1 | 1 | 0 | 1/3 | |
| | | | 7/7 = 1 | 3/3 = 1 | |

$2^3$ = 8 rows. → features.

Master table.

Combination ← $2^d$